Project Design Phase-II

Data Flow Diagram & User Stories

Date	25 September 2026
Team ID	23KD1A1507
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

The Automated Network Request Management system in ServiceNow is shown at two levels: Level 0 (context diagram) shows the system with its external entities, and Level 1 breaks it into its main processes and data stores.

DFD Level 0 - Context Diagram

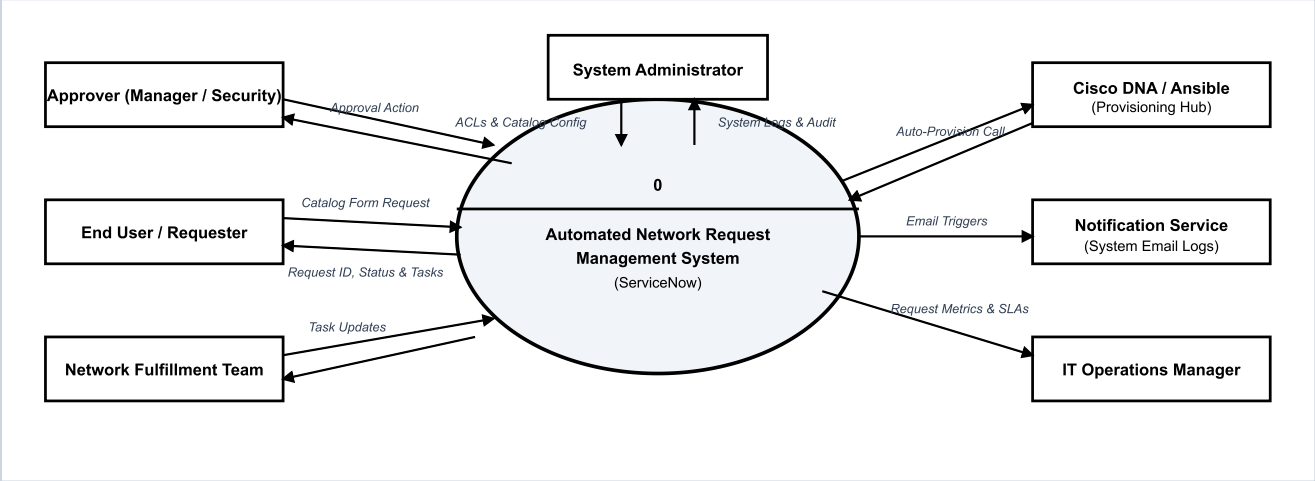


Figure 1: Data Flow Diagram - Level 0 (Context Diagram)

The Requester submits catalog items via the Service Portal (`/sp`); Flow Designer triggers automated approvals to Managers and Security teams; the Notification Service delivers email notifications; the Network Fulfillment Team executes port and connection tasks (`u_network_task`); and System Admins manage ACLs and tables (`u_database_table`).

DFD Level 1 - Part A: Intake, Approval, Execution & Notification

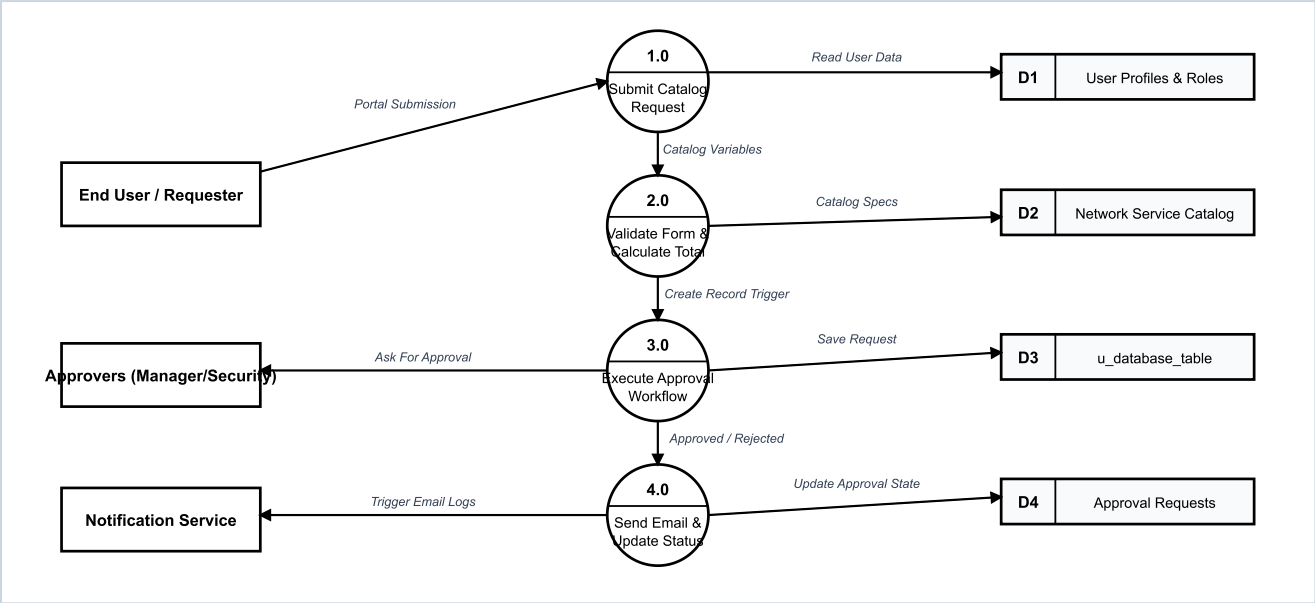


Figure 2: Data Flow Diagram - Level 1 (Processes 1.0 to 4.0)

ID	Process	Description
1.0	Submit Catalog Request	Captures request parameters from Service Portal form, authenticates requester details against D1, and initializes catalog transaction.
2.0	Validate Form & Variables	Applies dynamic Client Scripts and UI Policies, validates mandatory inputs, calculates total charges, and prepares payload.
3.0	Execute Approval Workflow	Flow Designer creates custom record in `u_database_table` (D3), checks rules, and sends Ask for Approval requests (D4) to managers/security teams.
4.0	Send Email & Update Record	Generates automated System Email Logs upon state change (Approved/Rejected), updates request status, and alerts the requester.

DFD Level 1 - Part B: Fulfillment, ACL Enforcement & Analytics



Figure 3: Data Flow Diagram - Level 1 (Processes 5.0 and 6.0)

ID	Name	Description
5.0	Fulfill Task & Enforce ACLs	Assigns fulfillment task (`u_network_task`) to network engineers, enforces field-level Access Control Rules (ACLs), and logs task completion.
6.0	Generate Reports & Analytics	Extracts metrics from `u_database_table` (D3) to display request volumes, fulfillment SLAs, and bottleneck analytics for management.
D1	User Profiles & Roles	Stores requester details, department roles, email addresses, and approval authority hierarchies.

ID	Name	Description
D2	Network Service Catalog	Contains catalog item definitions, variables sets, connection types, and pricing structures.
D3	u_database_table	Primary database table storing request record number, requester, connection type, mobile, amount, and status.
D4	Approval Requests	Tracks approval state (Requested / Approved / Rejected / Escalated) and approver comments.
D5	Network Task (`u_network_task`)	Fulfillment work tasks assigned to engineering teams for physical switch/port allocation.

User Stories

The user stories for the product are listed below by user type and functional requirement (epic) across development sprints.

User Type	Functional Req. (Epic)	Story ID	User Story / Task	Acceptance Criteria	Priority	Release
End User / Requester	Catalog & Form Submission	USN-1	As a requester, I can access the Network Request item in Service Portal (`/sp`).	Form loads with clear title and variable sets.	High	Sprint-1
		USN-2	As a requester, I can select connection type and input required contact details.	Dynamic fields show/hide based on selection.	High	Sprint-1
	Tracking & Alerts	USN-3	As a requester, I receive a confirmation email with a unique ticket auto-number.	Email log generated with valid record link.	High	Sprint-1
		USN-4	As a requester, I can track my ticket status in real-time on the portal.	Status updates reflect Flow Designer state.	Medium	Sprint-2

User Type	Functional Req. (Epic)	Story ID	User Story / Task	Acceptance Criteria	Priority	Release
Approver (Manager / Security)	Approval Workflow	USN-5	As a manager, I receive approval requests via email and ServiceNow portal.	One-click Approve / Reject actions are available.	High	Sprint-1
		USN-6	As a security officer, I am auto-assigned high-sensitivity access requests.	Flow Designer routes based on connection type.	High	Sprint-2
		USN-7	As an approver, I can add mandatory comments upon rejecting a request.	Comment field is enforced on rejection.	Medium	Sprint-2
Network Fulfillment Team	Task Execution & ACLs	USN-8	As a network engineer, I am automatically assigned fulfillment tasks (`u_network_task`).	Task contains all submitted catalog variables.	High	Sprint-1
		USN-9	As an engineer, I can update work notes and mark tasks as Complete.	Closing task updates parent ticket status.	High	Sprint-2
		USN-10	As a fulfiller, field-level ACLs restrict me from modifying requester contact info.	ACL rule blocks unauthorized write access.	Medium	Sprint-2
IT Operations Manager	Reports & Analytics	USN-11	As an ops manager, I can view real-time dashboards of pending vs closed requests.	Reports display counts per connection type.	Medium	Sprint-3
		USN-12	As an ops manager, I can track fulfillment SLA times across teams.	Analytics chart shows average resolution hours.	Low	Sprint-3

User Type	Functional Req. (Epic)	Story ID	User Story / Task	Acceptance Criteria	Priority	Release
System Administrator	Administration & Setup	USN-13	As an admin, I can configure custom tables (`u_database_table`) and auto-numbering.	Prefix and field structures update seamlessly.	High	Sprint-1
		USN-14	As an admin, I can test and execute Flow Designer logic triggers.	Execution details log steps without script errors.	High	Sprint-1